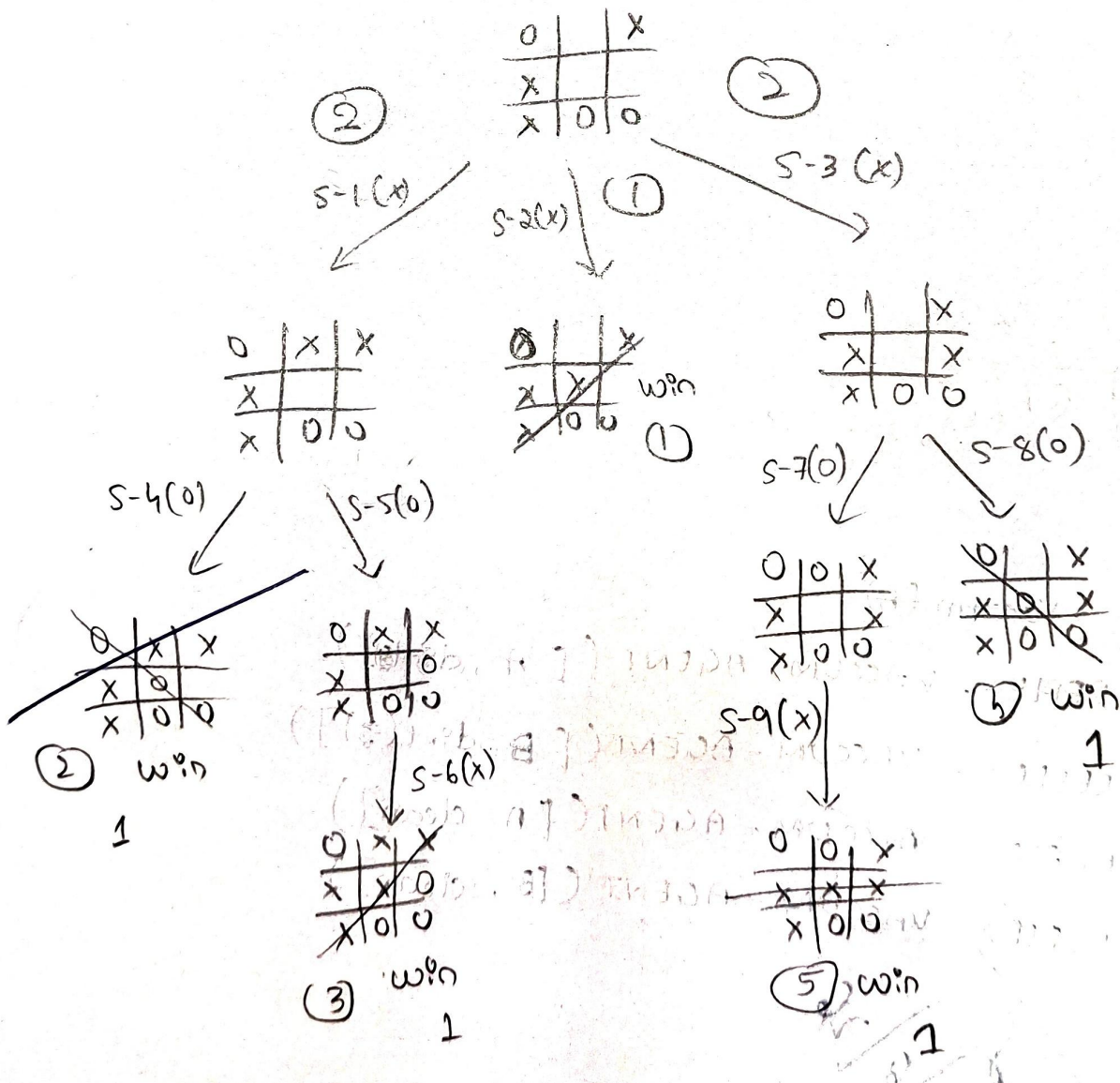


Tic - Tac - Toe. (Algorithm)

- step 1 : Initialize 3x3 board.
- step 2 : human play as O
- step 3 : Robot play as X
- step 4 : human places O in empty grid
- step 5 : Robot places X in empty grid
- step 6 : If any O or X comes in a same row, column or diagonal, return win
- step 7 : If all the empty places (grid) occupied, return draw



path cost : 2.

• Vacuum Cleaner

Function REFLEX-VACUUM-AGENT([location; status]) returns
an action.

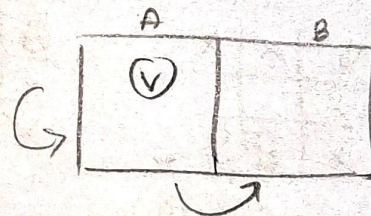
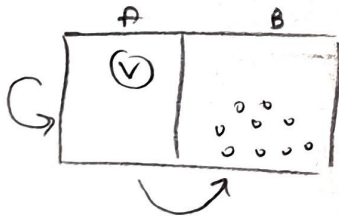
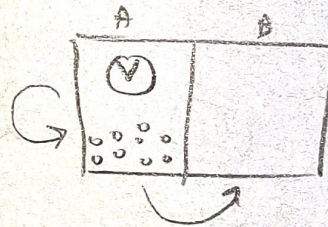
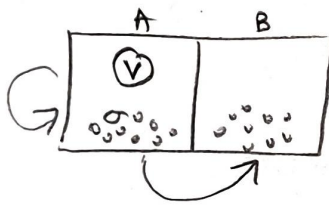
If status = Dirty then return Pick the dirt

else if location = A return Right

else if location = B then return Left

> If robot i.e. Vacuum cleaner in Loc A.

i.e. Initial state : location A.



if Vacuum (A)

REFLEX-VACUUM-AGENT([A, dirt])

REFLEX-VACUUM-AGENT([B, dirt(~~2~~)])

REFLEX-VACUUM-AGENT([A, clean])

REFLEX-VACUUM-AGENT([B, clean])

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Tic-Tac-Toe Output:

Welcome to Tic-Tac-Toe!

You are 'x' and the AI is 'o'

Choose a position (0-8): 0

x		

[0]

AI's turn...

x		
	o	

Choose a position (0-8): 2

x		x
	o	

AI's turn...

x	o	x
	o	

Choose a position (0-8): 7

x	o	x
	o	
	x	

AI's turn...

x	o	x
o	o	
	x	

choose a position (0-8): 5

x	o	x
o	o	x
	x	

AI's turn...

x	o	x
o	o	x
	x	o

choose a position (0-8): 6

x	o	x
o	o	x
x	x	o

It's a tie!

o/p :

0		x
x		
x	0	0

computer picks position: 1.

0	0	x
x		
x	0	0

Your turn(x) : 4

0	0	x
x	x	
x	0	0

computer picks position: 5

0	0	x
x	x	0
x	0	0

Game over ! Total path cost from Root: 2

Congratulations! You(x) win!

0		x
x		
x	0	0

S-1

0	0	x
x		
x	0	0

S-2

0	0	x
x	x	
x	0	0

win.

path cost : 2

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